

Coding & Solution

Date	01 July 2026
Team ID	SWTID-2026-7290
Project Name	Smart Lender
Maximum Marks	5 Marks

Solution Summary

Field	Details
Repository Link / URL	https://github.com/tejaswinimaddireddy678-web/AI-ML-and-GEN-AI-Track-Project-Template
Programming Language(s)	Python
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Key Features Implemented	1. Loan eligibility prediction using Machine Learning. 2. Data preprocessing (handling missing values and categorical encoding). 3. Exploratory Data Analysis (Univariate, Bivariate and Multivariate Analysis). 4. Multiple ML models (Decision Tree, Random Forest, KNN and XGBoost) were trained and evaluated. 5. Best-performing model selected for prediction. 6. Flask-based web application for loan eligibility prediction. 7. Model serialization using Pickle for deployment.
Pending / Incomplete Features	1. Advanced exception handling can be improved. 2. User authentication and database integration are not included. 3. Cloud deployment can be implemented in future versions.
Setup / Run Instructions	1. Clone the repository from GitHub. 2. Install the required Python libraries (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn and Flask). 3. Open the project folder and run the Jupyter Notebook to train the model if required. 4. Execute app.py to start the Flask application. 5. Open the generated local URL in a web browser. 6. Enter applicant details and click Predict to view the loan eligibility result.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Partial
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The Smart Lender project follows a structured machine learning workflow that includes data preprocessing, exploratory data analysis, model training, evaluation, and deployment using Flask. The code is organized into reusable modules with meaningful naming conventions, making it easier to understand and maintain. The application successfully predicts loan eligibility based on applicant details. Future improvements can include enhanced exception handling, database integration, and cloud deployment.